FIGURE 12: SCIENTIFIC EVIDENCE & VALIDATION TAXONOMY MATRIX

Deterministic Classification of System Parameters: Measured, Validated, Calculated, Simulated & Demonstrated

[MEASURED]

Direct Bench Instruments

- **Quiescent Current:** Pending PPK2 Measurement
- **DSP FFT Latency:** 2.49 ms (DWT Cycles)
- **MCU Die Temp:** 25.4°C–26.8°C (TEMP)
- **Hardware Weight:** 67 g (Digital Scale)
- **Enclosure Size:** 65x55x15 mm (Calipers)

[VALIDATED]

Empirical Calibrations

- **Brood Core RTD:** ±0.1°C NIST (TMP117)
- **Spatial Grid:** ±0.5°C (5x DS18B20)
- **Audio Stream:** 16 kHz 24-bit PCM
- **CUSUM Drift:** -0.02°C/hr Queenless
- **Model 2 AI:** Ensemble Classifier
- **Test Suite:** 27/27 Tests Pass

[CALCULATED]

Analytical Physics Models

- **Packet Airtime:** 77.06 ms (36B v2 Calc)
- **Daily Energy:** 0.85 mWh/day (15m)
- **LoRa LOS Range:** Modelled Free-Space
- **Link Margin:** 26.16 dB @ 1 km
- **Apiary Duty Cycle:** 0.202% (<1.0% Limit)
- **BOM Unit Cost:** $18.74 (No Custom PCB)

[SIMULATED]

Multiphysics Solvers

- **Antenna Return Loss:** S_11 = -22.4 dB (HFSS)
- **Antenna VSWR:** 1.16 : 1 (50 Ohm Match)
- **Radiation Eff.:** 91.4% (ABS Casing)
- **Thermal Core:** 34.5°C (Fluent CFD)
- **Convective Plume:** 0.18 m/s Natural Draft

[DEMONSTRATED]

Physical System Builds

- **Transmitter Node:** WisBlock RAK5005-O
- **Receiver Reader:** RPi 3B+ & Waveshare
- **Wiring Terminals:** 4:2 Levers & PG-7
- **Local Server:** FastAPI Daemon
- **Offline Database:** SQLite WAL Store